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شركة ايه زت تكنولوجي المحدودة

Web Service Description

For

AZ Payment Gateway

Version 0.2

Prepared by Software Development Department

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02-05-2024

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Revision History

Name	Reason for Changes	Date	Version
Hassan Elmukashfi	First Creation.	02-05-2024	0.1
Hassan Elmukashfi	Update third case	15-12-2024	0.2

1.0 Inroduction

1.1 Document Purpose

This document is intended to describe the web services provided by AZ Payment Gateway to perform specific transactions. This document also provides brief description of services available and a list of fields required for each service.

1.2 Glossary

JSON	Java Script Object Notation

2.0 Web Service Description

All transaction requests will be called by external APIs or mobile App to AZ payment Gateway.

2.1 Web Service Protocol

AZ payment Gateway web service is based on HTTP POST/GET with MIME type JSON.

2.2 Web Service Connection

AZ payment Gateway uses HTTP connection over secured VPN (firewall) connections.

Ex: http:// [url]:[port]/[api]/[api-version]/[service method]

3.0 Web Service Flow

3.1 Request Process

mobile should issue a first request to get all Categories available as start such that the list will only contains as:

- Category Name.
- Category ID.
- Category Image.

Response will only contain active categories, at request time any update done in back-end after the request should be inquiry again.

After displaying categories fetched from previous step and User select a category, the mobile will invoke the backend again to get All billers registered to selected Category i.e.

- If User selects Telecom, then backend should return all telecom companies available
- In sub-List User would expect to see either Zain, sudani or MTN.

When the mobile shows the available list of Billers (service providers) such that the user selects a certain provider, another request to list services available within the Biller (service provider) selected and will response as:

- Service ID
- Service Name
- Service Image

Response will only contain active services in biller, at request time any update done in back-end after the request should be inquiry again.

When the mobile’s user selects a certain service then a final request to shape the last page will be send with the following Parameters:

- Category ID.
- Biller ID.
- Service ID.
- Language.

Backend receives the above parameters and then start to prepare a parameter list that matches the paths selected in scenarios and returns data contains

Parameter	Description	Usage
OrderID	Contains a number that's describes its location within the mobile and its page	A mobile should start building the mobile page by this order.
Name	The name which will be shown to user i.e., Phone, Meter No	This will vary according to the language sent in request either Arabic -English or any specified language.
Key	This is the key sent to Payment Gateway API in the Json object	
regex	This contains regular expression that the mobile should match the values against it before sending to backend	Mobile and backend uses this regular expression to validate data in a runtime or request since no other validator are available
inputType	Mobile uses this value which will be shared between mobile and backend to specify how this field will be rendered	This value either: • inputText • menu (dropdown list) • checkbox • textArea • radio buttons
options	Used when the above value is menu (drop-down list) or radio buttons	This specifies the multiple values of the input. Backend sends this as key-value pairs.
minValue	Minimum value of this input Used only if value is accepting numbers	For amount and ranges If (-1) then it's any.
maxValue	maxValue of an input used if value accepting numbers	For amount and ranges If (-1) then it's any.
Length	Specify the length input and used to limit user input	-1 for data doesn't obey this rule such as data with options or textArea.

3.2 Generating Request

In previous steps the mobile user reaches the point when he will submit the request and mobile should construct the request as follow steps:

- Service ID: collected when the user specifies the required service.
- Language: mobile context language.
- Biller ID: biller selected in scenario #2.
- RequestParams: this will contain a list of request parameters of the service specified above and will be containing data as follow:
    - Key: a json object parameter which value corresponds to value of Key value.
    - Value: a json object parameter which value will hold user input or choice.
- User Data: user data object contains user id, user account and other information needed.

If above step was successful and request data was generated the mobile invoke a route for performing inquiry or payment.
Backend receives request and then perform mapping and generate service specific request, the response value will be one of three.

3.3 Service's Response

3.3.1 #1

This service Performs both inquiry and payment in one step thus no other process is performed in the mobile and then mobile should only show the receipt. Back end will send data as follow:

- **Response Code**: API indicator wither the process was successful or not. Default value is zero which means successful.
- **Response Message**: message shows either the error and will be in multilanguage support depending on locale.
- **Response_params**: list of json objects contain:
        - Key
        - Value
        - Display Name
- **Final_status**: this property will be 0 if the process flow finish else will contains the next service ID that will be sent to API to make the next process
- **Request_params**: will be null indicating the process is finished.

Mobile should check both **response code** and **Final_status** parameters to decide showing receipt or error message.

3.3.2 #2

This service Performs inquiry first as a sperate request then it will be two steps process and Back end will send data with changes in Request_params and final_status parameters:

- **Request_params**: will be having data with structure described in previous section.
- **Final_status**: will contains value except 0, which will be the next Service ID in Payment phase.

Mobile should generate a new page that shows data received from process in Response_params and prompt user to input other data specified in Request_params received in response. Mobile should always check the parameter **isPayment** whether it's true or false to prompt Account selection for payment.

3.3.3 #3

The mobile captures in response the value **isPayment** is **true** then mobile should prompt user to input Account which payment will be through and generate request as below:

- Service ID: collected when the user specifies the required service.
- Language: mobile context language.
- Biller ID: biller selected in scenario #2.
- RequestParams: this will contain a list of request parameters of the service specified above and will be containing data as follow:
  - Key: a json object parameter which value corresponds to value of Key value.
  - Value: a json object parameter which value will hold user input or choice.
- User Data: user data object contains user id, user account and other information needed.

Back end performs its process and then return data as described in scenario #2 with changes in:

- **Final_status**: will be 0.
- **Request_params**: will be empty or null.
- **Response_params**: will contains all data to be displayed with its order ID.